

- 5 *BRCA2* is a tumour-suppressor gene. Its gene product, BRCA2, is involved in DNA repair. If the DNA cannot be repaired, BRCA2 has a role in causing the cell to die. BRCA2 is found in cells of breast tissue.

When a mutation occurs in *BRCA2*, damaged DNA may not be repaired and this increases the risk of breast cancer.

When a double-stranded piece of DNA breaks, BRCA2 binds to the damaged DNA directly and interacts with the enzyme RAD51 to repair the damage.

Repairing DNA prevents other mutations and gene rearrangements from occurring which could otherwise lead to breast cancer.

- (a) Double-stranded DNA breaks occur naturally during meiosis.

State the event that is initiated as a result of double-stranded breaks during meiosis.

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- (b) Scientists have identified hundreds of mutations in *BRCA2*, but not all of these mutations will increase the risk of cancer. One specific mutation in *BRCA2*, known as 999del5, is found in 0.6% of the general global population.

Iceland is an island country in the North Atlantic Ocean. The ancestors of most of the current population are people who arrived to settle in Iceland in AD 874.

In the Icelandic population, mutation 999del5 is the cause of 7–8% of breast cancer cases in women and 40% of breast cancer cases in men. This is much higher than the percentage of breast cancer cases in the general global population.

- (i) Suggest **and** explain how mutation 999del5 accounts for a very high percentage of breast cancer cases in Iceland compared with the general global population.

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- (ii) One of the largest global genetic screening programmes for breast cancer involves identifying people with mutations in *BRCA2*.

Outline the advantages of genetic screening for mutations in *BRCA2*.

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- (iii) Suggest **one** advantage to a country of a genetic screening programme for breast cancer that screens for specific mutations in *BRCA2* in the population.

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- (c) *Lipocalin 2* (*Lcn2*) is a cancer-promoting gene (oncogene). When *Lcn2* is expressed, it can result in breast cancer.

Research is being carried out to see if gene editing of *Lcn2* could be used to treat breast cancer.

Gene editing was used to treat human cancer cells that had been implanted into mice to form a tumour. The treatment stopped *Lcn2* from being expressed in the cancer cells and resulted in a significant reduction in the growth of the tumour. There was no negative effect in normal tissues.

- (i) Suggest how DNA editing stopped *Lcn2* from being expressed in cancer cells.

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- (ii) Suggest why the expression of **only** *Lcn2* was affected.

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[Total: 11]